

Specification for Approval

Customer	
Product Name	Wire Wound Molded SMD Power Inductors
Customer P/N:	
Cjiang P/N:	FTC Series

[☒ New Released, ☐ Revised]

SPEC No.:

REMARK:

Customer Approval Feedback

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1. Scope

Features

- 1.1 Metal material for large current and low loss.
- 1.2 High performance (Isat) realized by metal dust core.
- 1.3 Low loss realized with low Rdc.
- 1.4 Closed magnetic circuit design reduces leakage flux.
- 1.5 Vinyl thermal spray, better surface compactness.
- 1.6 100% lead (Pb) free meet RoHS standard.



Application

- 2.1 DC/DC converters.
- 2.2 Pad, Smartphone.
- 2.3 Portable gaming devices, Smart wear, Wi-Fi module.
- 2.4 Notebooks, VR, AR.
- 2.5 LCD displays, HDDs, DVCs, DSCs, etc.
- 2.6 Baseband power supply, Amplifier, Power management, Module power supply, Camera power manageme.

2. Ordering Procedure

FTC 2016 10 S 1R0 M B C *

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

①Series Name: Mini Molding Power Inductors

②External Dimensions(L×W):2016=2.0*1.6 mm

③External Dimensions(H):10=1.0 mm

④Size Tolerance:S=±0.2mm D=±0.1mm

⑤Inductance value:1R0=1.0uH

⑥Tolerance:K=±10% M=±20% N=±30%

⑦Coating color:B=Black G=Gray

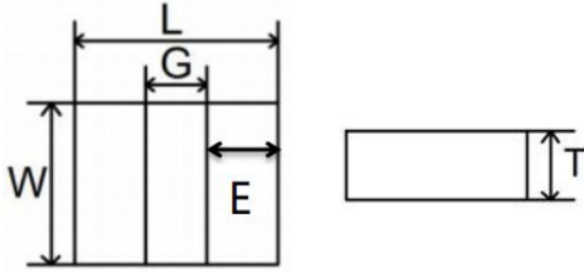
⑧Product type:C=Common

⑨Special define:A=Routine B~Z=Special

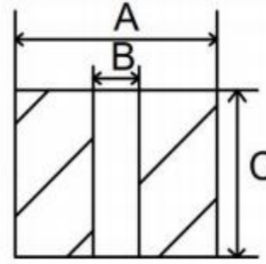
For special characteristics, please refer to the specific values in Item 5 "Specifications".

3. SHAPE AND DIMENSIONS

Outline Dimensions



Recommend Land Pattern Dimensions



Units:mm

Series	L	G/Typ	W	E	T	A/Typ	B/Typ	C/Typ
FTC100505S	1.0±0.15	0.4	0.5±0.15	0.30±0.2	0.50Max.	1.20	0.30	0.60
FTC100506S	1.0±0.15	0.4	0.5±0.15	0.30±0.2	0.60Max.	1.20	0.30	0.60
FTC100565S	1.0±0.15	0.4	0.5±0.15	0.30±0.2	0.65Max.	1.20	0.30	0.60
FTC100765D	1.0±0.1	0.4	0.7±0.1	0.30±0.2	0.65Max.	1.10	0.30	0.80
FTC100708D	1.0±0.1	0.4	0.7±0.1	0.30±0.2	0.80Max.	1.10	0.30	0.80
FTC121065S	1.2±0.2	0.4	1.0±0.2	0.40±0.2	0.65Max.	1.30	0.30	1.10
FTC160806S	1.6±0.2	0.6	0.8±0.2	0.50±0.2	0.60Max.	1.70	0.50	0.90
FTC160865S	1.6±0.2	0.6	0.8±0.2	0.50±0.2	0.65Max.	1.70	0.50	0.90
FTC160808S	1.6±0.2	0.6	0.8±0.2	0.50±0.2	0.80Max.	1.70	0.50	0.90
FTC141265S	1.4±0.2	0.5	1.2±0.2	0.45±0.2	0.65Max.	1.50	0.40	1.30
FTC141207S	1.4±0.2	0.5	1.2±0.2	0.45±0.2	0.70Max.	1.50	0.40	1.30
FTC141208S	1.4±0.2	0.5	1.2±0.2	0.45±0.2	0.80Max.	1.50	0.40	1.30
FTC201206S	2.0±0.2	0.6	1.2±0.2	0.70±0.2	0.60Max.	2.10	0.50	1.30
FTC201265S	2.0±0.2	0.6	1.2±0.2	0.70±0.2	0.65Max.	2.10	0.50	1.30
FTC201208S	2.0±0.2	0.6	1.2±0.2	0.70±0.2	0.80Max.	2.10	0.50	1.30
FTC201210S	2.0±0.2	0.6	1.2±0.2	0.70±0.2	1.00Max.	2.10	0.50	1.30
FTC201212S	2.0±0.2	0.6	1.2±0.2	0.70±0.2	1.20Max.	2.10	0.50	1.30
FTC201655S	2.0±0.2	0.6	1.6±0.2	0.70±0.2	0.55Max.	2.10	0.50	1.70
FTC201665S	2.0±0.2	0.6	1.6±0.2	0.70±0.2	0.65Max.	2.10	0.50	1.70
FTC201607S	2.0±0.2	0.6	1.6±0.2	0.70±0.2	0.70Max.	2.10	0.50	1.70
FTC201608S	2.0±0.2	0.6	1.6±0.2	0.70±0.2	0.80Max.	2.10	0.50	1.70
FTC201610S	2.0±0.2	0.6	1.6±0.2	0.70±0.2	1.00Max.	2.10	0.50	1.70
FTC201610D	2.0±0.1	0.6	1.6±0.1	0.70±0.2	1.00Max.	2.10	0.50	1.70
FTC201612S	2.0±0.2	0.6	1.6±0.2	0.70±0.2	1.20Max.	2.10	0.50	1.70
FTC252055S	2.5±0.2	0.8	2.0±0.2	0.85±0.2	0.55Max.	2.60	0.70	2.10
FTC252075S	2.5±0.2	0.8	2.0±0.2	0.85±0.2	0.75Max.	2.60	0.70	2.10
FTC252008S	2.5±0.2	0.8	2.0±0.2	0.85±0.2	0.80Max.	2.60	0.70	2.10

FTC252010S	2.5±0.2	0.8	2.0±0.2	0.85±0.2	1.00Max.	2.60	0.70	2.10
FTC252012S	2.5±0.2	0.8	2.0±0.2	0.85±0.2	1.20Max.	2.60	0.70	2.10
FTC322510S	3.2±0.2	1.1	2.5±0.2	1.05±0.2	1.00Max.	3.20	1.00	2.50
FTC322512S	3.2±0.2	1.1	2.5±0.2	1.05±0.2	1.20Max.	3.20	1.00	2.50
FTC322520S	3.2±0.2	1.1	2.5±0.2	1.05±0.2	2.00Max.	3.20	1.00	2.50
FTC303010D	3.0±0.1	1.0	3.0±0.1	1.00±0.2	1.00Max.	3.00	0.90	3.00
FTC303012D	3.0±0.1	1.0	3.0±0.1	1.00±0.2	1.20Max.	3.00	0.90	3.00
FTC303015D	3.0±0.1	1.0	3.0±0.1	1.00±0.2	1.50Max.	3.00	0.90	3.00
FTC303018D	3.0±0.1	1.0	3.0±0.1	1.00±0.2	1.80Max.	3.00	0.90	3.00
FTC303020D	3.0±0.1	1.0	3.0±0.1	1.00±0.2	2.00Max.	3.00	0.90	3.00
FTC404008S	4.1±0.2	1.4	4.1±0.2	1.35±0.2	0.80Max.	4.10	1.30	4.10
FTC404010S	4.1±0.2	1.4	4.1±0.2	1.35±0.2	1.00Max.	4.10	1.30	4.10
FTC404012S	4.1±0.2	1.4	4.1±0.2	1.35±0.2	1.20Max.	4.10	1.30	4.10
FTC404015S	4.1±0.2	1.4	4.1±0.2	1.35±0.2	1.50Max.	4.10	1.30	4.10
FTC404020S	4.1±0.2	1.4	4.1±0.2	1.35±0.2	2.00Max.	4.10	1.30	4.10
FTC404030S	4.1±0.2	1.4	4.1±0.2	1.35±0.2	3.00Max.	4.10	1.30	4.10

4. Marking

No Marking

5. Specifications

1>1005 Series

FTC100505(1.0*0.5*0.50mm)

P/N	L0(μH) @(0A) 1MHz	Rdc(mΩ)		Heat rating current Irms(A)		Saturation current Isat(A)	
		Typical	Max	Typical	Max	Typical	Max
FTC100505SR47MBCA	0.47	190	220	1.3	1.2	1.5	1.3

FTC100506(1.0*0.5*0.60mm)

P/N	L0(μH) @(0A) 1MHz	Rdc(mΩ)		Heat rating current Irms(A)		Saturation current Isat(A)	
		Typical	Max	Typical	Max	Typical	Max
FTC100506SR47MBCA	0.47	160	190	1.5	1.4	1.65	1.55
FTC100506S1R0MBCA	1.0	280	320	1.0	0.8	1.15	1.05
FTC100506S1R0MBCD	1.0	270	300	1.0	0.8	1.15	1.05

FTC100565(1.0*0.5*0.65mm)

P/N	L0(μH) @(0A) 1MHz	Rdc(mΩ)		Heat rating current Irms(A)		Saturation current Isat(A)	
		Typical	Max	Typical	Max	Typical	Max
FTC100565SR47MBCA	0.47	150	175	1.7	1.6	1.85	1.75

2>1007 Series

FTC100765(1.0*0.7*0.65mm)

P/N	L0(μH)	Rdc(mΩ)	Heat rating current Irms(A)	Saturation current Isat(A)
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FTC252012S(2.5*2.0*1.2mm)

P/N	L0(μH) @(0A) 1MHz	Rdc(mΩ)		Heat rating current Irms(A)		Saturation current Isat(A)	
		Typical	Max	Typical	Max	Typical	Max
FTC252012SR10MBCA	0.10	6.0	10	12	10.5	13.5	12.5
FTC252012SR10MBCD	0.10	4.0	4.8	16.0	15.0	15.0	14.0
FTC252012SR11MBCA	0.11	4.2	5.0	15.0	14.0	14.0	13.0
FTC252012SR15MBCA	0.15	7.0	11	11.5	10	13.0	12.0
FTC252012SR15MBCD	0.15	5.7	7.0	11.5	10	13.0	12.0
FTC252012SR22MBCA	0.22	9.0	14	8.2	7.6	9.6	9.0
FTC252012SR22MBCD	0.22	8.0	10	11	8.0	12	10
FTC252012SR22MBCF	0.22	6.0	8.0	9.5	9.0	11	10
FTC252012SR24MBCA	0.24	10	15	8.0	7.5	9.3	8.8
FTC252012SR24MGCA	0.24	10	15	8.0	7.5	9.3	8.8
FTC252012SR24MBCD	0.24	6.5	8.5	9.5	9.0	10.0	9.5
FTC252012SR24MBCF	0.24	8.7	10	8.0	7.5	11	10
FTC252012SR24MBCE	0.24	8.0	11	8.0	7.5	9.3	8.8
FTC252012SR33MBCA	0.33	11	17	6.8	6.4	8.3	7.8
FTC252012SR33MBCD	0.33	10	12	9.0	8.5	9.0	8.5
FTC252012SR33MBCF	0.33	7.5	9.5	9.0	8.5	9.0	8.5
FTC252012SR47MBCA	0.47	13	19	6.5	6.0	7.5	7.0
FTC252012SR47MBCD	0.47	11	13	8.0	7.5	8.5	8.0
FTC252012SR47MGCD	0.47	11	13	8.0	7.5	8.5	8.0
FTC252012SR56MBCA	0.56	16	19	6.5	6.0	7.1	6.6
FTC252012SR68MBCA	0.68	17	23	6.3	5.5	6.5	6.0
FTC252012SR68MBCD	0.68	15	18	7.5	7.0	6.7	6.0
FTC252012SR82MBCA	0.82	19	24	5.8	5.3	6.5	5.8
FTC252012SR82MBCD	0.82	19	23	6.3	5.8	6.5	5.8
FTC252012S1R0MBCA	1.0	35	42	5.3	4.8	5.6	5.0
FTC252012S1R0MBCD	1.0	16	22	6.1	5.6	6.5	6.0
FTC252012S1R0MGCD	1.0	16	22	6.1	5.6	6.5	6.0
FTC252012S1R0MBCF	1.0	30	36	5.5	5.0	5.6	5.0
FTC252012S1R2MBCA	1.2	40	45	3.8	3.4	4.5	4.1
FTC252012S1R2MBCD	1.2	29	35	5.7	5.2	5.3	4.8
FTC252012S1R5MBCA	1.5	44	50	3.7	3.2	4.5	4.1
FTC252012S1R5MGCA	1.5	44	50	3.7	3.2	4.5	4.1
FTC252012S1R5MBCF	1.5	31	35	4.5	4.1	4.5	4.2
FTC252012S1R5MBCD	1.5	27	32	4.6	4.2	4.7	4.4

FTC252012S1R5MBCB	1.5	46	53	4.6	4.2	5.2	4.7
FTC252012S2R2MBCA	2.2	55	65	3.0	2.7	3.8	3.3
FTC252012S2R2MGCA	2.2	55	65	3.0	2.7	3.8	3.3
FTC252012S2R2MBCD	2.2	52	60	3.9	3.5	4.1	3.7
FTC252012S3R3MBCA	3.3	80	97	2.8	2.5	3.0	2.7
FTC252012S4R7MBCA	4.7	150	170	2.2	1.9	2.4	2.1
FTC252012S4R7MGCA	4.7	150	170	2.2	1.9	2.4	2.1
FTC252012S4R7MBCB	4.7	150	170	2.4	2.2	2.7	2.5
FTC252012S6R8MBCB	6.8	245	270	2.1	2.0	2.2	2.1
FTC252012S6R8MBCA	6.8	245	270	1.8	1.6	2.0	1.7
FTC252012S8R2MBCA	8.2	250	285	1.6	1.4	1.8	1.55
FTC252012S100MBCA	10.0	330	400	1.2	1.05	1.6	1.45
FTC252012S100MGCA	10.0	330	400	1.2	1.05	1.6	1.45
FTC252012S150MBCA	15.0	500	565	1.4	1.3	1.4	1.3
FTC252012S220MBCA	22.0	740	800	1.2	1.1	1.1	1.0

9>3225Series

FTC322510S(3.2*2.5*1.0mm)

P/N	L0(μH) @(0A) 1MHz	Rdc(mΩ)		Heat rating current Irms(A)		Saturation current Isat(A)	
		Typical	Max	Typical	Max	Typical	Max
FTC322510SR22MBCA	0.22	9.0	11	8.5	8.0	8.5	8.0
FTC322510SR33MBCA	0.33	11	15	8.3	7.8	8.3	7.8
FTC322510SR47MBCA	0.47	17	22	6.4	5.9	8.3	7.6
FTC322510SR68MBCA	0.68	22	28	6.2	5.7	7.5	7.0
FTC322510S1R0MBCA	1.0	25	30	5.4	4.9	6.0	5.3
FTC322510S1R0MBCD	1.0	21	25	5.5	5.0	5.6	5.1
FTC322510S1R5MBCA	1.5	34	42	4.0	3.6	5.0	4.4
FTC322510S2R2MBCA	2.2	55	66	3.7	3.4	4.0	3.5
FTC322510S3R3MBCA	3.3	105	120	2.7	2.3	3.7	3.3
FTC322510S4R7MBCA	4.7	125	140	2.3	1.9	2.8	2.5
FTC322510S6R8MBCA	6.8	290	320	1.9	1.6	2.4	2.0
FTC322510S100MBCA	10.0	325	365	2.2	1.8	2.2	1.8

Test remarks

Note 1.: All test data is referenced to 25 °C ambient.

Note 2.: Test Condition:1MHz, 1.0Vrms.

Note 3.: Irms:DC current (A) that will cause an approximate ΔT of 40 °C.

Note 4.: Isat:DC current (A) that will cause L0 to drop approximately 30%.

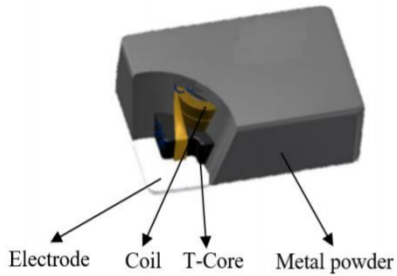
Note 5.: Operating Temperature Range -55°C to + 125°C.

Note 6.: The part temperature (ambient + temp rise) should not exceed 125 under °C the worst case operating conditions. Circuit design, component placement, PCB trace size and thickness, airflow and other cooling provision all affect the part temperature. Part temperature should be verified in the end application.

Note 7.: The rated current as listed is either the saturation current or the heating current depending on which value is lower.

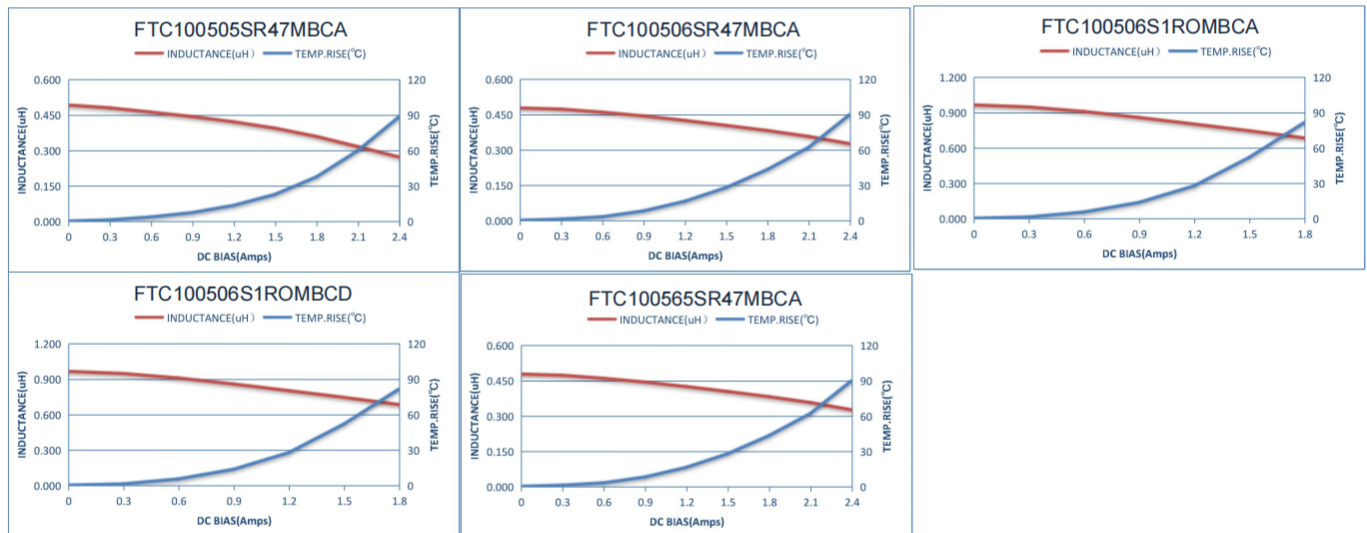
Note 8.withstand Dc voltage : 20[v]

6. Structure

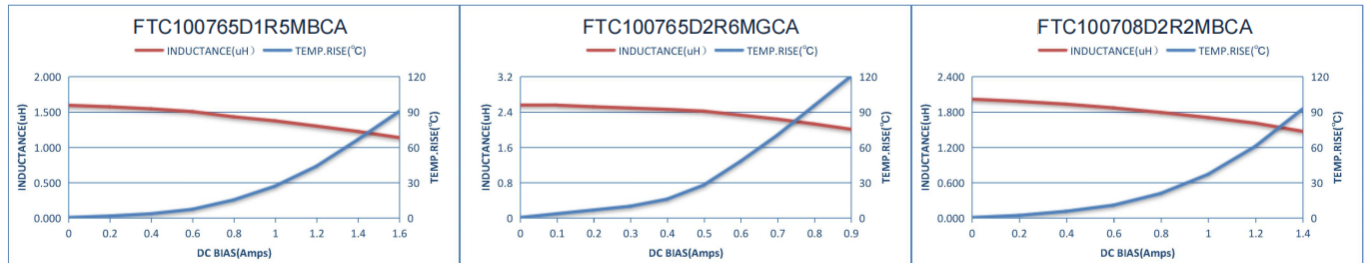


7. Current Characteristic

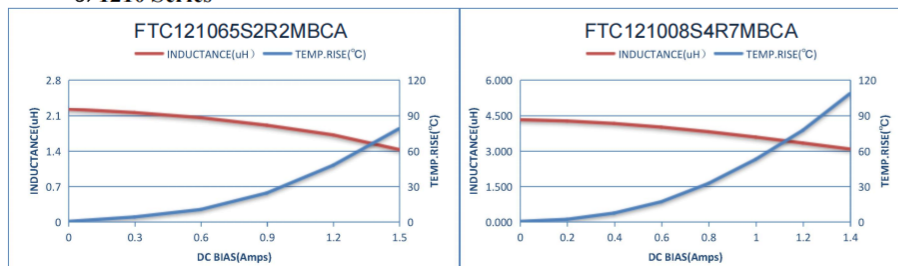
1>1005Series

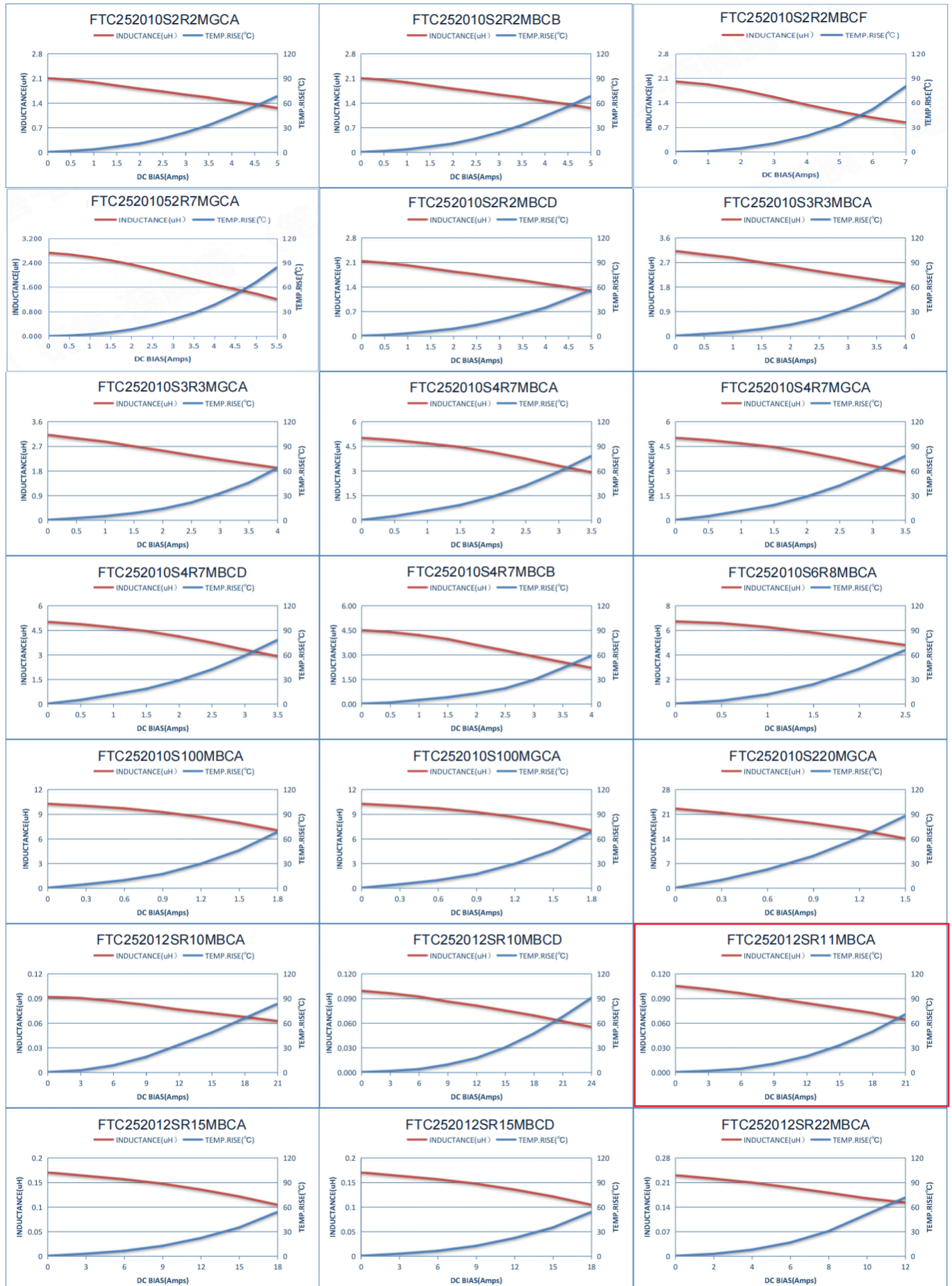


2>1007 Series

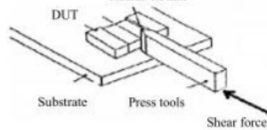
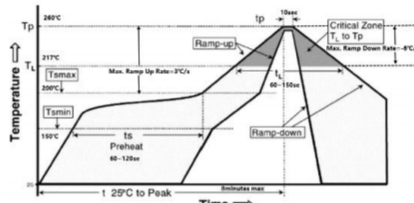
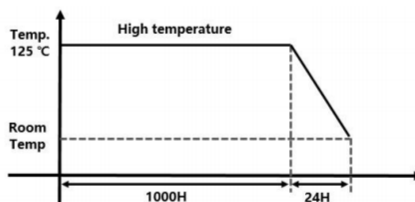
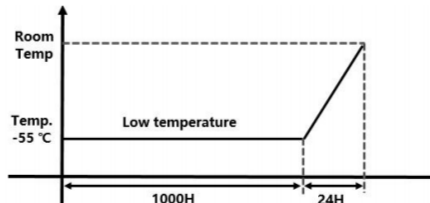


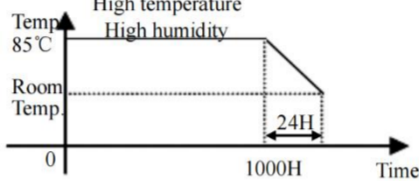
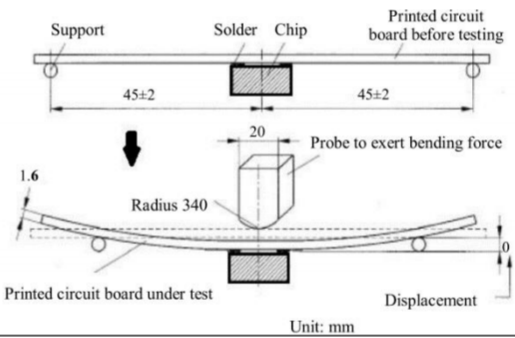
3>1210 Series





8. Reliability

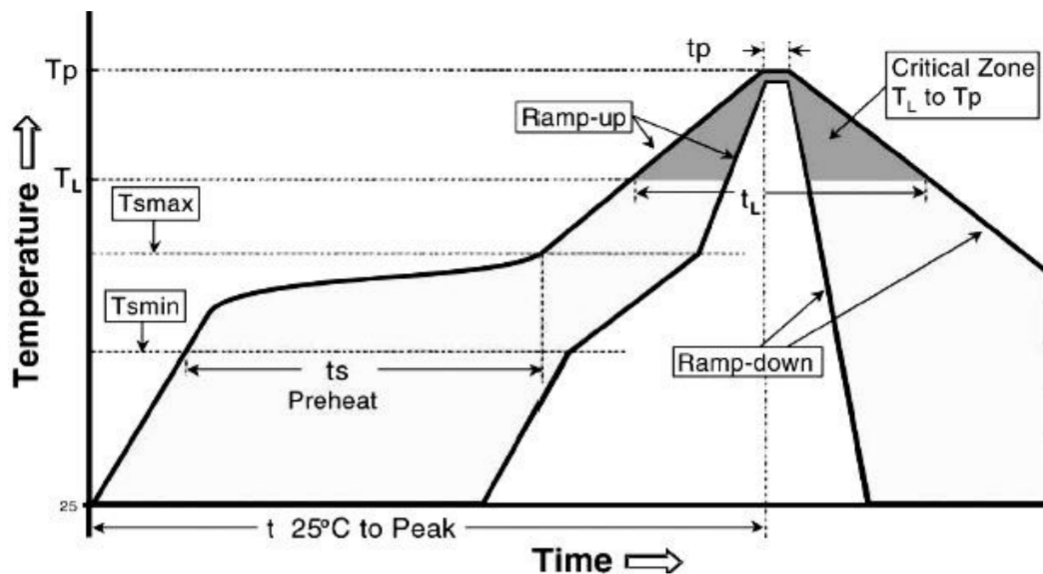
No.	Item	Requirements	Test Methods and Remarks	Reference	Sample Size								
1	Solderability	(1) No case deformation or change in appearance. (2) Terminal area must have 95% min. Solder coverage.	①Temperature:245± 5°C. ②Solder Composition: Sn/Ag3.0/Cu0.5(Pb-Free). ③Sample immersion tin furnace 5 ±0.5s.	AEC-Q200 (J-STD 002)	32								
2	Adhesion of teral electrode	(1)Strong bond between the pad and the core, without come off PCB.	①Preconditioning: 245℃Reflow 3 times ②Inductors shall be subjected to (260+0/-5°C.)℃ for (10±5)s Soldering in the base whit 0.3mm solder. ③Aplombelectrode way plus tax X N for (10±1) seconds. <table border="1"><tr><td>series</td><td>"X" N</td></tr><tr><td>1008</td><td>6</td></tr><tr><td>1210~1608</td><td>8</td></tr><tr><td>2012</td><td>12</td></tr></table> 	series	"X" N	1008	6	1210~1608	8	2012	12	AEC-Q200 (AEC-Q200-006)	32
series	"X" N												
1008	6												
1210~1608	8												
2012	12												
3	Reflow test	(1) No physical damage. (2) ΔL0/L0 ≤10%	① The peak temperature: 260+0/-5°C. ② Reflow:3times. ③ Temperature curve is as below 	AEC-Q200 (MIL-STD-202 Method 210)	32								
4	High temperature	(1) No physical damage. (2) ΔL0/L0 ≤10%	① Preconditioning: Bake at 125+5°C for 24±0.5H, 245°C Reflow 3 times ②Temperature: 125±2°C. ③ Time : 1000 hours. ④Measurement at 24±4 hours after test conclusion 	AEC-Q200 (MIL-STD -202 Method 108)	77								
5	Low temperature	(1) No physical damage. (2) ΔL0/L0 ≤10%	①Preconditioning: Bake at 125+5°C for 24±0.5H, 245°C Reflow 3 times ②Temperature: -55±2°C. ③Time : 1000 hours. ④ Measurement at 24±4 hours after test conclusion 	JESD22-A119A	77								
6	Thermal shock	(1) No physical damage. (2) ΔL0/L0 ≤10%	①Preconditioning: Bake at 125+5°C for 24±0.5H, 245°C Reflow 3 times ②Repeat 500 cycle as follow : (-55±2 °C ,30±3minutes) 、 (Room temperature, 5 minutes) 、 (+125±2 °C ,30±3minutes) 、 (Room temperature, 5 minutes) ③Measurement at 24±4 hours after test conclusion	MIL-STD -202 Method 107	77								

No.	Item	Requirements	Test Methods and Remarks	Reference	Sample Size
7	Resistance to Soldering Heat	(1) No physical damage. (2) $ \Delta L_0/L_0 \leq 10\%$	①Solder Composition: Sn/Ag3.0/Cu0.5(Pb-Free). ②Solder Temperature: 260±5°C. ③Immersion Time: 10±1sec.	AEC-Q200 (MIL-STD-202 Method 210)	32
8	Static Humidity	(1) No physical damage. (2) $ \Delta L_0/L_0 \leq 10\%$	①Preconditioning: Bake at 125±5°C for 24±0.5H, 245°C Reflow 3 times ②1000 hours, 85°C/85%RH. ③ Unpowered. ④Measurement at 24±4 hours after test conclusion 	AEC-Q200 (MIL-STD-202 Method 103)	77
9	Board Flex	(1) No physical damage. (2) $ \Delta L_0/L_0 \leq 10\%$	①Preconditioning: 245°C Reflow 3 times ②Part mounted on a 100mm*40mm FR4 PCB board, which is 1.6±0.2 mm thick and as a Layer-thickness 35 μm ± 10 μm. ③ Bending speed is 1mm/s. ④ Keeping the P.C Board 2 mm minimum for 60 seconds. 	AEC-Q200 (AEC-Q200-005)	30
10	Vibration	(1) No physical damage. (2) $ \Delta L_0/L_0 \leq 10\%$	①Preconditioning: 245°C Reflow 3 times ②Frequency range : 10~2000Hz. ③Amplitude: 1.5mm or 20g. ④ Sweep time and duration: 10~2000~10Hz for 20 minutes. ⑤ Each four hours in X,Y,Z direction, 12hours in total.	AEC-Q200 (MIL-STD-202 Method 204)	32
11	Mechanical Shock	(1) No physical damage. (2) $ \Delta L_0/L_0 \leq 10\%$	① Preconditioning: 245°C Reflow 3 times ②Peak acceleration:100G/S ③Duration of pulse:6ms ④3times in each of 6(±X, ±Y, ±Z) axes.	AEC-Q200 (MIL-STD-202 Method 213)	32
12	Loading at High Temperature	(1) No physical damage. (2) $ \Delta L_0/L_0 \leq 10\%$	①Preconditioning: Bake at 125±5°C for 24±0.5H, 245°C Reflow 3 times ②Temperature: 85±2°C. ③ Time : 1000 hours. ④ Applied Current : Rated current. ⑤Measurement at 24±4 hours after test conclusion	AEC-Q200 (MIL-STD-202 Method 108)	77

9. Soldering Condition

(This is for recommendation, please customer perform adjustment according to actual application)

Recommend Reflow Soldering Profile : (solder : Sn96.5 / Ag3 / Cu0.5)



Profile Feature	Lead (Pb)-Free solder
Preheat:	
Temperature Min (T _{smin})	150℃
Temperature Max (T _{smax})	200℃
Time (T _{smin} to T _{smax}) (ts)	60 -120 seconds
Average ramp-up rate: (T _{smax} to T _p)	3℃ / second max.
Time maintained above :	
Temperature (T _L)	217℃
Time (t _L)	60-150 seconds
Peak Temperature (T _p)	260℃
Time within $\begin{smallmatrix} +0 \\ -5 \end{smallmatrix}$ ℃ of actual peak Temperature (tp) ²	10 seconds
Ramp-down Rate	6℃/second max.
Time 25℃ to Peak Temperature	8minutes max.

Allowed Re-flow times : 2 times

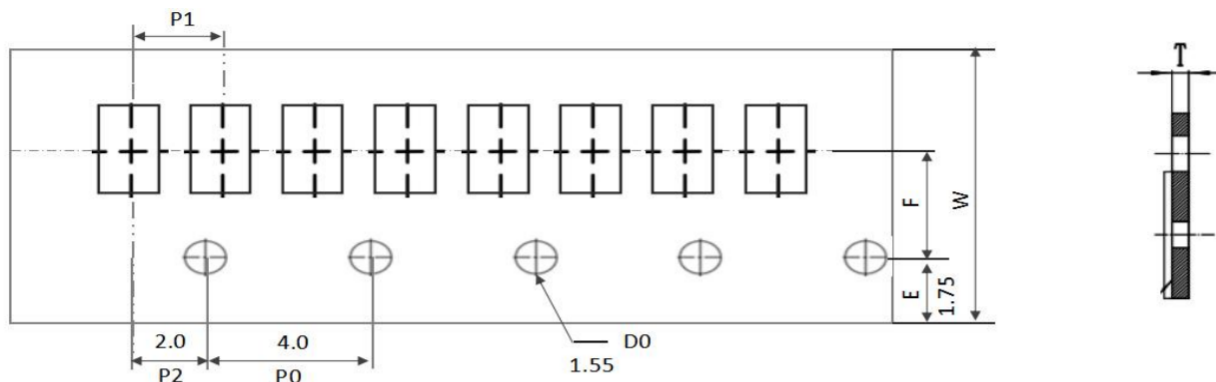
Remark : To avoid discoloration phenomena of chip on terminal electrodes, please use N₂ Re-flow furnace .

10. Packing

10.1 Dimension of plastic taping: (Unit: mm)

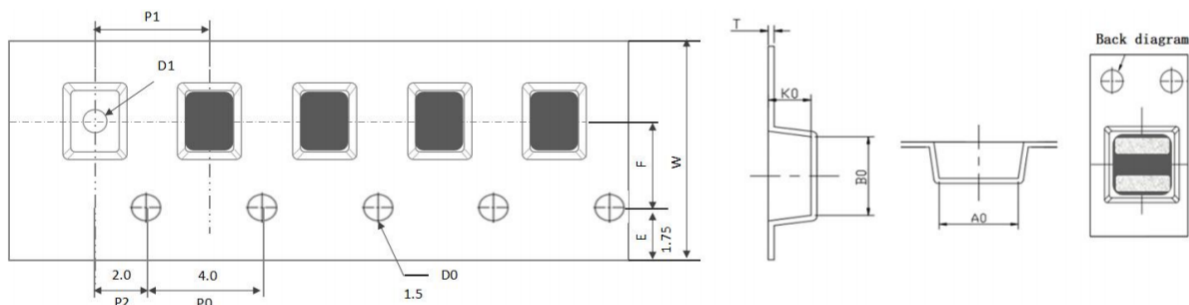
The following dimensions are related to the actual fit of the machine, for reference only.

10.1.1 1005-1007 Series



Series	W	A0	B0	D0	F	P1	T	Packing Quantity
Tolerance	± 0.10	± 0.04	± 0.04	± 0.20	± 0.10	± 0.05	± 0.05	
100506/100505	8.0	0.75	1.25	1.55	3.5	2.0	0.65	5K
100565	8.0	0.75	1.25	1.55	3.5	2.0	0.75	
100765	8.0	0.90	1.20	1.55	3.5	2.0	0.75	
100708	8.0	0.90	1.20	1.55	3.5	2.0	0.95	

10.1.2 1210-3030 Series



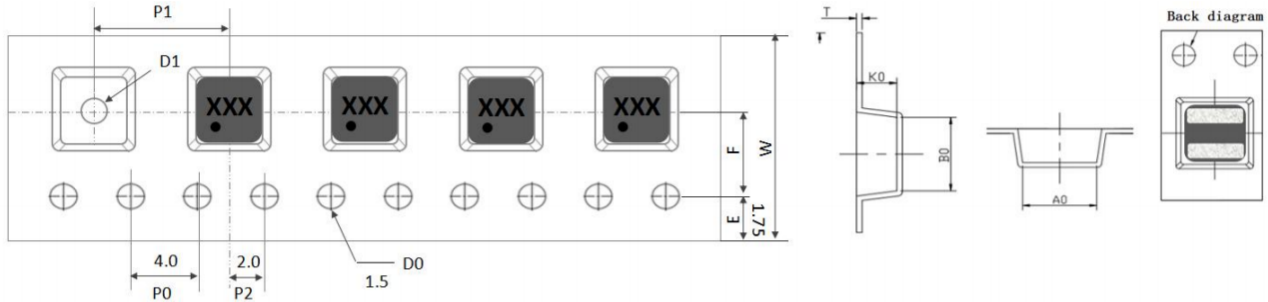
Series	W	A0	B0	D1	F	K0	P1	T	Packing Quantity
Tolerance	/	/	/	± 0.20	± 0.10	/	± 0.10	± 0.05	
121065	8.0 ± 0.10	1.30 ± 0.05	1.50 ± 0.05	0.7	3.5	0.80 ± 0.10	4.0	0.22	3K
160806/160865	8.0 ± 0.10	1.00 ± 0.05	1.82 ± 0.05	0.6	3.5	0.80 ± 0.10	4.0	0.22	
160808	8.0 ± 0.10	1.04 ± 0.05	1.82 ± 0.05	0.6	3.5	0.95 ± 0.05	4.0	0.22	
141265	8.0 ± 0.10	$1.50 \pm 0.10 / -0.05$	$1.70 \pm 0.10 / -0.05$	0.6	3.5	0.80 ± 0.10	4.0	0.22	
141207/141208	8.0 ± 0.10	$1.50 \pm 0.10 / -0.05$	$1.70 \pm 0.10 / -0.05$	0.6	3.5	0.95 ± 0.10	4.0	0.22	
201206/201265	8.0 ± 0.10	$1.40 \pm 0.10 / -0.05$	$2.25 \pm 0.10 / -0.05$	1.0	3.5	0.80 ± 0.10	4.0	0.25	
201208	8.0 ± 0.10	1.50 ± 0.10	2.30 ± 0.10	1.0	3.5	1.00 ± 0.10	4.0	0.22	
201210	8.0 ± 0.10	1.50 ± 0.10	2.35 ± 0.10	1.0	3.5	1.20 ± 0.10	4.0	0.22	
201212	8.0 ± 0.10	1.50 ± 0.10	2.35 ± 0.10	1.0	3.5	1.40 ± 0.10	4.0	0.22	
201655/201665	8.0 ± 0.10	1.95 ± 0.10	2.45 ± 0.10	1.0	3.5	0.80 ± 0.10	4.0	0.25	
201607/201608	8.0 ± 0.10	1.90 ± 0.10	2.35 ± 0.10	1.0	3.5	1.00 ± 0.10	4.0	0.25	
201610	8.0 ± 0.10	1.95 ± 0.10	2.35 ± 0.10	1.0	3.5	1.15 ± 0.10	4.0	0.25	
201612	8.0 ± 0.10	1.90 ± 0.10	2.30 ± 0.10	1.0	3.5	1.35 ± 0.10	4.0	0.25	
252055	8.0 ± 0.10	$2.35 \pm 0.10 / -0.05$	$2.80 \pm 0.10 / -0.05$	1.0	3.5	0.80 ± 0.10	4.0	0.23	
252075/252008	8.0 ± 0.10	$2.35 \pm 0.10 / -0.05$	$2.80 \pm 0.10 / -0.05$	1.0	3.5	1.00 ± 0.10	4.0	0.23	

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252010	8.0±0.10	2.45±0.10	2.80±0.10	1.0	3.5	1.20±0.10	4.0	0.25	
252012	8.0±0.10	2.35±0.10/-0.05	2.80±0.10/-0.05	1.0	3.5	1.35±0.10	4.0	0.23	
322510	8.0±0.10	2.80±0.10/-0.05	3.50±0.10/-0.05	1.0	3.5	1.15±0.10	4.0	0.25	
322512	8.0±0.10	2.90±0.10/-0.05	3.50±0.10/-0.05	1.0	3.5	1.35±0.10	4.0	0.25	
322520	8.0±0.10	2.90±0.10/-0.05	3.50±0.10/-0.05	1.0	3.5	2.20±0.10	4.0	0.23	
303010/303012	12.0±0.30	3.40±0.10	3.40±0.10	1.5	5.5	1.40±0.10	8.0	0.30	2K
303015	12.0±0.30	3.40±0.10	3.40±0.10	1.5	5.5	1.70±0.10	8.0	0.30	
303018	12.0±0.30	3.25±0.10	3.25±0.10	1.5	5.5	2.00±0.10	8.0	0.23	
303020	12.0±0.30	3.40±0.10	3.45±0.10	1.5	5.5	2.20±0.10	8.0	0.30	

10.1.3 4040 Series



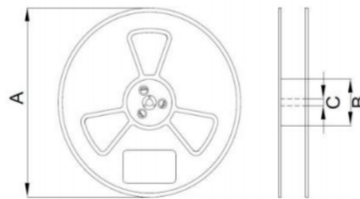
Series	W	A0	B0	D1	F	K0	P1	T	Packing Quantit
Tolerance	/	/	/	±0.20	±0.10	/	±0.10	±0.05	
404008	12.0±0.30	4.40±0.10	4.40±0.10	1.5	5.5	1.00±0.10	8.0	0.30	3K
404010/404012	12.0±0.30	4.40±0.10	4.40±0.10	1.5	5.5	1.30±0.10	8.0	0.30	
404015	12.0±0.30	4.40±0.10	4.40±0.10	1.5	5.5	1.70±0.10	8.0	0.30	
404020	12.0±0.30	4.40±0.10	4.40±0.10	1.5	5.5	2.20±0.10	8.0	0.30	
404030	12.0±0.30	4.40±0.10	4.40±0.10	1.5	5.5	3.10±0.10	8.0	0.35	2K

10.2 Dimension of Reel : (Unit: mm)

Packing size "W=8mm" using 178mm Reel
Packing size "W>12mm" using 330mm Reel

Type	A ±2.0	B ±2.0	C ±2.0
All	178	60	13.0

Type	A ±2.0	B ±2.0	C ±2.0
All	330	100	13.0



11. Note

11.1 Storage Period

To maintain the solderability of terminal electrodes and to keep the packing material in good condition, product should be used within 12 months from the time of delivery. And the solderability of products electrodes may decrease as time passes, so in case of storage over 12 months, solderability shall be checked before actual usage.

11.2 Storage Conditions

Store products in a warehouse in compliance with the following condition:

Temperature: Inductors (product with taping) +15 to +35℃; Humidity: 25~70%RH.

If the product is not used up 24H after opening, desiccant should be placed in the package and sealed

11.3 Do not subject products to rapid changes in temperature and humidity.

11.4 Do not store the products in chemical atmosphere such as one containing sulfurous acid gas or alkaline gas, that will causes poor solderability and corrosion of inductors.

11.5 Do not store products in bulk packaging to prevent collision among inductors which causes core chipping and wire breakage.

11.6 Store products on pallets to protect from humidity, dust, etc.

11.7 Avoid heat shock, vibration, direct sunlight, etc.

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11.8 When designing the PCB, please consider the installation position of the non-magnetic shielded components to avoid failures caused by magnetic interference.

11.9 Do not place this product near magnets or objects with magnetic force.

11.10 The product will self-heat (temperature rise) due to power-on, and sufficient margin should be left in thermal design.

11.11 Mounting density

If this product is placed near heat-generating products, be sure to implement sufficient heat-dissipating measures. If this product is subjected to a significant amount of heat from other products, this could adversely affect product quality, resulting in a circuit malfunction or failure of the mounted section. Also, be sure that the product is used in a manner so that the heat that the product is subjected to from other products does not exceed the upper limit of the rated operating temperature for the product.

11.12 Since the static electricity carried by the human body will be transmitted to the ground wire, please use an anti-static wrist strap.

11.13 Grease on human hands may lead to decreased solderability. Please avoid direct contact with the terminals.

11.14 This product refers to the general standard used in audio-visual entertainment, home appliances, computers, office automation, communications, power modules, LED lighting, measuring equipment. Machine tools, industrial control panels and other general electronic equipment. And the general electronic equipment should be used under the usual operation and usage methods.

11.15 When this product is used in occasions other than general electronic equipment, such as: Automotive Electronic products, medical equipment, military equipment, Aerospace equipment, submarine equipment, etc. please be sure to contact the company's sales department, the company will cooperate with customer needs, and negotiate different intended use described in this product

11.16 Reworking with soldering iron

ITEM	Requirement
Pre-heating	150° C/approx. 1 min
Tip temperature of soldering iron	350° C max.
Soldering time	3 s (+1 s, -0 s)
Number of reworking operations	1 times max.
It is recommended to replace the product directly, rework may cause poor appearance	

11.17 We cannot warrant against failure caused by any use of our product that deviates from the intended use as described in this product specification.

11.18 Please approve our product specifications or transact the approval sheet for product specifications before ordering.

11.19 Our specification limits the quality of the component as a single unit. Please ensure the component is thoroughly evaluated in your application circuit.

12. Record

Version	Description	Page	Date
D6	newly increased: FTC201208S100MBCA FTC201606S2R2MBCA FTC201606S4R7MBCA FTC201610S220MBCA FTC252012SR22MBCF FTC303012D3R3MBCA FTC404030S470MBCA FTC201212SR10MBCA FTC252008SR33MBCA FTC201212SR68MBCA Revise: FTC201610S1R0MBCD Irms from 4.6/4.2 to 4.8/4.3 FTC201610S1R0MGCD Irms from 4.6/4.2 to 4.8/4.3 FTC201610S1R0MBCD Isat from 4.7/4.2 to 4.8/4.2 FTC201610S1R0MGCD Isat from 4.7/4.2 to 4.8/4.2 Delete: FTC252075S100MGCA	1~48	May.30.2025
D7	newly increased: FTC121008S4R7MBCA FTC201206SR47MBCA FTC201210S1R2MBCA FTC201212S2R2MBCA FTC252012SR24MBCF FTC252012SR24MBCE FTC322520SR22MBCA FTC322520S100MBCA FTC303015DR22MBCA FTC404030S150MBCA	1~51	Jun.30.2025

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